Software Engineering Course Project

Small Mart Automation

Project Specification and Planning Document

Cloud-Based Point-of-Sale, Inventory, Employee, Loyalty, Analytics and Loss-Prevention System

Project designed by: _____

Team members: _____

Course / Section: _____

This project develops an integrated software system for small-to-medium retail marts. The system replaces fragmented manual records with a shared platform that coordinates checkout, inventory, employees, customers and management decision support.

Manual Small-Mart Operation	Automated Operation	Business Outcomes
• Paper / spreadsheet stock records	• Barcode / SKU-based POS	• Faster checkout
• Separate cash register records	• Real-time inventory deduction	• Fewer stockouts
• Manual stock counts	• Low-stock + expiry alerts	• Less waste
• Limited expiry visibility	• Supplier purchase orders	• Better records
• Manual reordering	• Employee activity tracking	• Better decisions
• Weak shrinkage visibility	• Analytics + loss-prevention alerts	

Figure 1: From manual small-mart operation to integrated automation.

Project emphasis: Inventory Management is the main coordination challenge. The system is designed to reduce stockouts and overstock while connecting inventory data to sales, reordering, analytics and loss prevention.

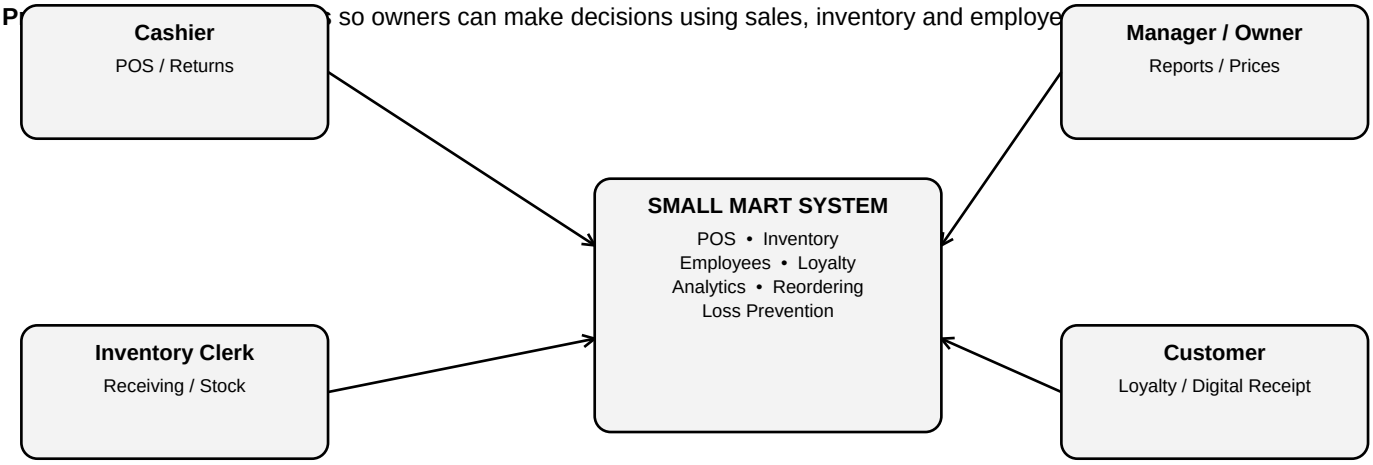
1. Project Description

Small retail marts frequently depend on paper records, basic cash registers or disconnected spreadsheets. These approaches make it difficult to know the true stock position, identify shrinkage, recognize fast-moving products, manage expiry dates and react quickly during busy checkout periods. The proposed Small Mart Automation system provides one integrated platform for daily operations and management decision support.

The primary goal is to automate the flow of information from the moment an item is received into inventory, through shelf stock and checkout, to reporting and replenishment. A sale immediately reduces available stock; low-stock thresholds can trigger reorder recommendations; and managers can review revenue, margins, dead stock, employee activity and inventory discrepancies from a central dashboard.

The project targets four measurable business outcomes:

- **Reduce checkout time** by using barcode scanning, fast item lookup and minimal-click cashier workflows.
- **Minimize out-of-stock incidents** through real-time inventory balances, thresholds and reorder support.
- **Lower operating costs and waste** by identifying dead stock, expiry risks and unnecessary overstock.
- **P** so owners can make decisions using sales, inventory and employe



Real-time shared data store / cloud application with offline-capable POS

Figure 2: High-level interaction between users and the Small Mart Automation system.

1.1 Statement of Requirements

The system shall support role-based access so that each user sees only the functions appropriate to their job. The primary roles are Manager/Owner, Cashier, Inventory Clerk/Stocker and Customer.

Role	Primary responsibilities
Manager / Owner	View analytics and financial reports; manage employees; set prices; manage suppliers; approve or review purchase orders; m
Cashier	Scan items or enter SKUs; apply valid discounts/taxes; process cash, card and mobile-wallet payments; issue receipts; proces
Inventory Clerk / Stocker	Receive shipments; scan products; update quantities; record batches and expiry dates; perform stock checks; resolve discrepa
Customer	Use self-checkout or optional QR/BYOD shopping workflow; earn and redeem loyalty points; access digital receipts and purcha

1.2 Core Functional Requirements

Module	Required capabilities
Point of Sale (POS)	Barcode scanning or manual SKU entry; real-time price lookup; tax and discount calculation; cash, credit/debit and mobile
Inventory Management	Real-time stock deduction; low-stock thresholds; reorder alerts; supplier records; batch numbers; expiry-date alerts; receiv
Employee Management	Clock-in/clock-out; shift scheduling; task assignment; employee profiles; cashier sales tracking; role-based permissions.
Customer Loyalty / CRM	Loyalty points; purchase history; digital receipts; optional targeted promotions. This module is treated as an extension if pr
Analytics / Decision Support	Daily revenue; top-selling items; peak hours; revenue-versus-cost analysis; profit-margin support; dead-stock reporting; op
Loss Prevention	Track voids, manual price overrides and inventory discrepancies; provide manager alerts when suspicious activity exceeds
Automated Reordering	Generate reorder recommendations or purchase orders when an SKU reaches its configured minimum stock level.

1.3 Inventory-Centered Coordination

Inventory management is the coordination hub for this project. A successful sale, shipment receipt, stock adjustment, return or expiry event can change the inventory state. POS therefore consumes current product and price data, while automated reordering and analytics consume inventory history. This focus allows the team to demonstrate a coherent software-engineering problem rather than building unrelated modules.

The core inventory workflow is:

- Receive shipment → record supplier, batch and expiry information.
- Put stock into inventory → update available quantity.
- Sell item → deduct stock in real time.
- Monitor threshold → flag low stock or expiry risk.
- Generate reorder → prepare purchase order/recommendation for manager review.
- Reconcile → compare system quantity with physical count and record discrepancy.

1.4 Non-Functional Requirements

- **Performance:** Common cashier operations should require minimal clicks and provide fast item lookup.
- **Availability:** Core POS functions should continue during temporary internet outages and synchronize when connectivity returns.
- **Security:** Role-based access, authenticated staff accounts, protected payment information and auditable privileged actions.
- **Usability:** Touchscreen-friendly interfaces, clear status indicators and consistent workflows.
- **Maintainability:** Modular architecture separating POS, inventory, employee, customer and analytics concerns.
- **Data integrity:** Sales and inventory updates should be transactional so that a completed sale does not silently leave stock unchanged.

1.5 System Design and Hardware Considerations

The proposed implementation is a cloud-based web application with an offline-capable POS layer. The cloud service provides a shared source of business data, while local POS devices can temporarily queue transactions during an outage and synchronize when the connection is restored.

Device / approach	Advantages	Disadvantages	Proposed use
Fixed POS terminal	Stable, secure, lower handling risk; familiar checkout setup.	Can create lines if too few terminals are available.	Primary cashier workstation.
Mobile handheld / tablet	Real-time shelf scanning; useful for receiving, stocktaking and price checks.	Battery, damage and loss concerns.	Inventory clerk / stocker.
Self-checkout kiosk	Can reduce staffed checkout workload.	Higher setup cost and greater shrinkage-control challenge.	Optional later expansion / future deployment.
Customer smartphone / BYOD	Enables QR shopping cart and digital checkout.	Requires customer adoption and stronger fraud controls.	Advanced extension.

1.6 User Interface and Authentication

The cashier interface should prioritize speed. A compact list or grid of frequently sold products (for example, milk, bread and other high-volume items) can supplement barcode scanning. The manager interface should emphasize a dashboard containing revenue, top-selling products, inventory warnings and peak-hour information. The inventory interface should be optimized for one-handed or touchscreen scanning.

Staff authentication can use a touchscreen PIN, swipe/badge card or barcode staff ID. Regardless of the physical mechanism, authorization is determined by the user's role. Sensitive actions such as manual price overrides, voids and inventory adjustments should create an auditable record.

1.7 Proposed Data Model

Entity	Representative attributes / relationships
Product	SKU, barcode, name, category, price, tax class, reorder threshold, supplier.
Inventory	SKU, quantity on hand, batch number, expiry date, storage location, adjustment history.
Employee	Employee ID, name, role, schedule, clock events, permissions.
Sale / SaleItem	Sale ID, cashier, timestamp, payment type, line items, discounts, tax, total.
Supplier / PurchaseOrder	Supplier information, ordered items, quantities, expected delivery, status.
Customer / LoyaltyAccount	Customer ID, contact identifier, points balance, permitted purchase history.

1.8 Domain Fieldwork

The team should visit at least one local small or medium-sized mart and interview an owner, manager or cashier. The purpose is not simply to confirm that automation is useful, but to identify the operational bottleneck that the software should address first. The findings should be used to justify design decisions in the final report.

Suggested interview questions

- What is the biggest frustration with the current POS or inventory system?
- How do you know when an item is running out?
- How often do stock counts disagree with the system or shelf quantity?
- How are expiry dates and perishable products tracked?
- What happens when the internet or POS system goes down?
- Which cashier actions require manager approval?
- How are supplier orders created and followed up?
- Which reports are most useful when deciding what to reorder or discontinue?

Fieldwork status: The questions above are part of the project plan. Actual interview findings should be inserted after the team completes the field visit; no customer or employee observations should be presented as facts before they are collected.

1.9 Competitive Analysis

System	Useful reference features	Project opportunity
Square POS	POS, payments, product and sales management.	Emphasize inventory coordination, expiry/batch tracking and small-mart shrinkag
Clover	POS hardware ecosystem, employee and sales features.	Provide a focused workflow for inventory-first small retail operations.
Loyverse	Retail POS, inventory and loyalty-oriented functions.	Differentiate through deeper reorder logic, discrepancy tracking and integrated d

1.10 Economic Feasibility

The feasibility analysis should compare software and hardware costs with the operational value produced by fewer stockouts, reduced waste, faster checkout, lower manual record-keeping effort and earlier detection of shrinkage. The team should avoid claiming a specific financial return until field data or defensible assumptions are collected.

A simple evaluation model is: **Net benefit = estimated annual savings + measurable operational benefit** – **annual system cost**. The team can then test conservative, expected and optimistic scenarios.

2. Extensions

The core system should be completed before advanced features are attempted. Extensions are selected to reinforce the inventory-centered problem rather than distract from it.

2.1 BYOD and QR-Code Shopping

Customers may scan a QR code when entering the store to create a digital shopping cart. As they shop, they scan product barcodes with their phone. The system records the cart and can support payment without a traditional checkout queue. Because this approach creates opportunities for unscanned or incorrectly scanned products, it should be treated as an extension and paired with receipt verification and exception handling.

2.2 Self-Checkout

A self-checkout interface can reuse the POS and inventory services. It should support barcode scanning, payment, receipt generation and controlled exception handling. Loss-prevention events such as repeated voids or unusual quantity changes should be logged for later review.

2.3 Advanced Analytics

The analytics layer can be expanded to include sales forecasting, peak-hour prediction, product-margin analysis and dead-stock reporting over 30/60/90-day windows. Forecasts should be clearly identified as predictions rather than guaranteed outcomes.

2.4 Automated Reordering

When a SKU reaches its reorder point, the system can generate a purchase-order recommendation. The manager may approve, edit or reject the recommendation. Future versions could incorporate supplier lead time, historical sales velocity and seasonality rather than relying only on a fixed threshold.

2.5 Other Extensions

- Multi-location support with consolidated inventory and sales reporting.
- Supplier performance tracking based on delivery time and order accuracy.
- Shelf-scanning workflows for periodic cycle counts.
- Customer-specific promotions based on consented purchase history.
- Advanced anomaly detection for potential shrinkage.

3. Technical Documentation and Software-Engineering Artifacts

The final project report should document the system using standard software-engineering artifacts. The artifacts should agree with one another: names used in the use cases should match the class model, database entities and implementation modules.

Artifact	Required content
Use Case Diagram	Manager views reports/manages employees/prices/orders; Cashier processes sales/returns; Inventory Clerk receives stock/
Class Diagram	Product, InventoryItem/Batch, Sale, SaleItem, Employee, Supplier, PurchaseOrder, Customer/LoyaltyAccount and supportin
Sequence Diagram	Example: cashier scans item → POS requests product/price → inventory validates availability → sale is recorded → invento
ERD / Database Schema	Products, inventory/batches, employees, sales/sale items, suppliers/purchase orders and customer/loyalty data with primary
Architecture Diagram	Web client/POS and handheld interfaces connected to application services, database, authentication and synchronization/of

3.1 Testing Strategy

- **Unit testing:** product lookup, tax/discount calculations, inventory deduction, threshold calculations and permission checks.
- **Integration testing:** POS-to-inventory updates, purchase-order creation, employee activity records and loyalty transactions.
- **System testing:** complete checkout, receiving, return, stock adjustment and reporting workflows.
- **Offline testing:** simulate internet loss during checkout, queue transactions locally, restore connectivity and verify synchronization.
- **User acceptance testing:** use realistic cashier and stock-clerk scenarios and record defects and usability observations.

3.2 Success Measures

The team should evaluate whether the prototype can demonstrate faster checkout workflows, accurate real-time stock deduction, useful low-stock/expiry alerts, reliable transaction records and manager-visible analytics. The final evaluation should report measured results from test scenarios rather than unsupported claims.

4. Project Schedule — Gantt / Timeline

The project schedule follows a work-breakdown approach. Each major activity has a defined duration, and activities that do not depend on one another may proceed in parallel. Milestones mark important baselines or completion points. The schedule is synchronized with the PERT network on the following page.

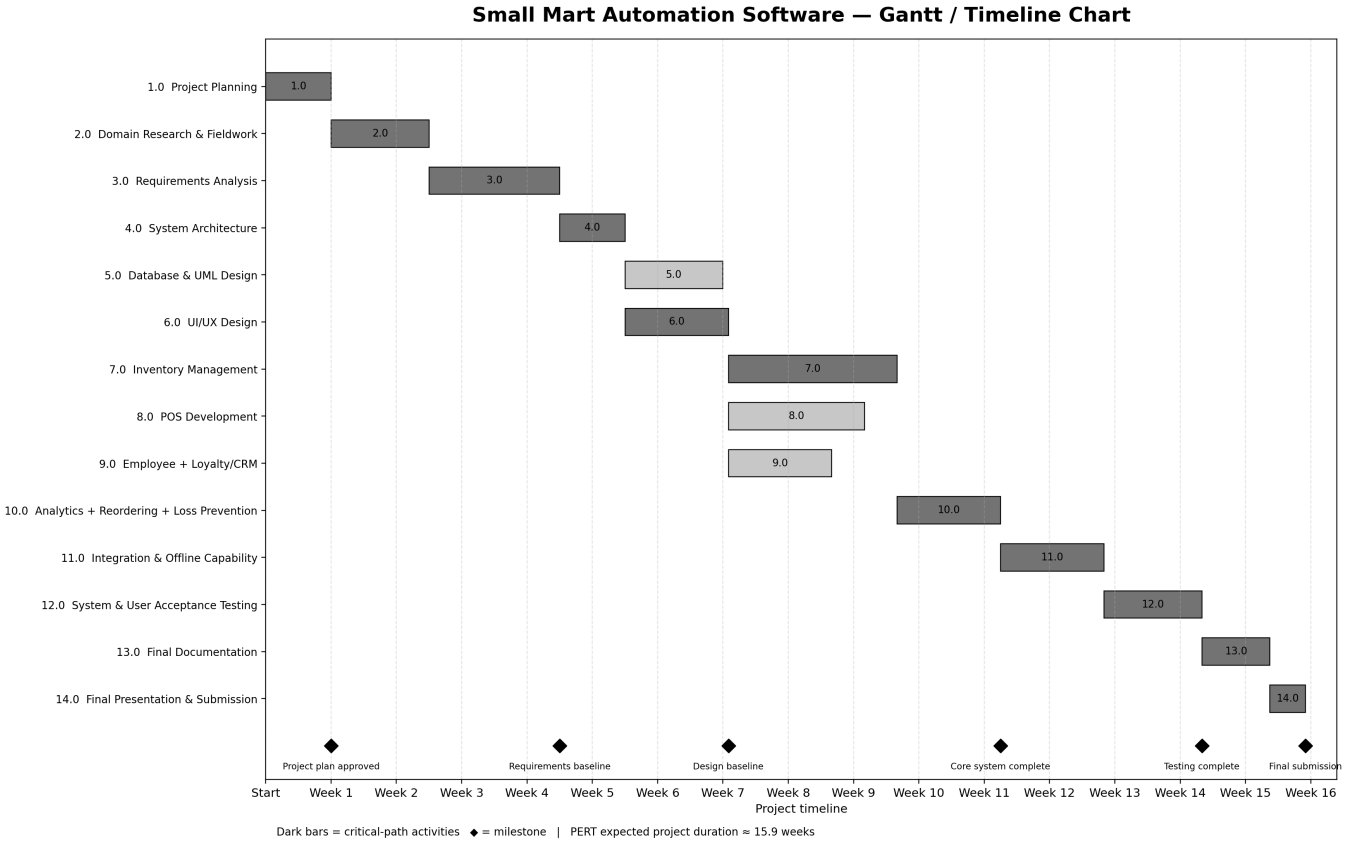


Figure 3: Small Mart Automation project Gantt/timeline chart. Dark bars identify activities on the calculated critical path; diamonds indicate milestones.

4.1 Major Milestones

Milestone	Target point
Project plan approved	End of Week 1
Requirements baseline	End of Week 4.5
Design baseline	About Week 7
Core system complete	About Week 11.25
Testing complete	About Week 14.33
Final submission	About Week 15.92

5. PERT / Task Network

PERT is used here to represent dependencies and estimate activity durations using optimistic (O), most likely (M) and pessimistic (P) values. The expected time is calculated as TE = (O + 4M + P) / 6. The longest dependency path is treated as the critical path for planning purposes.

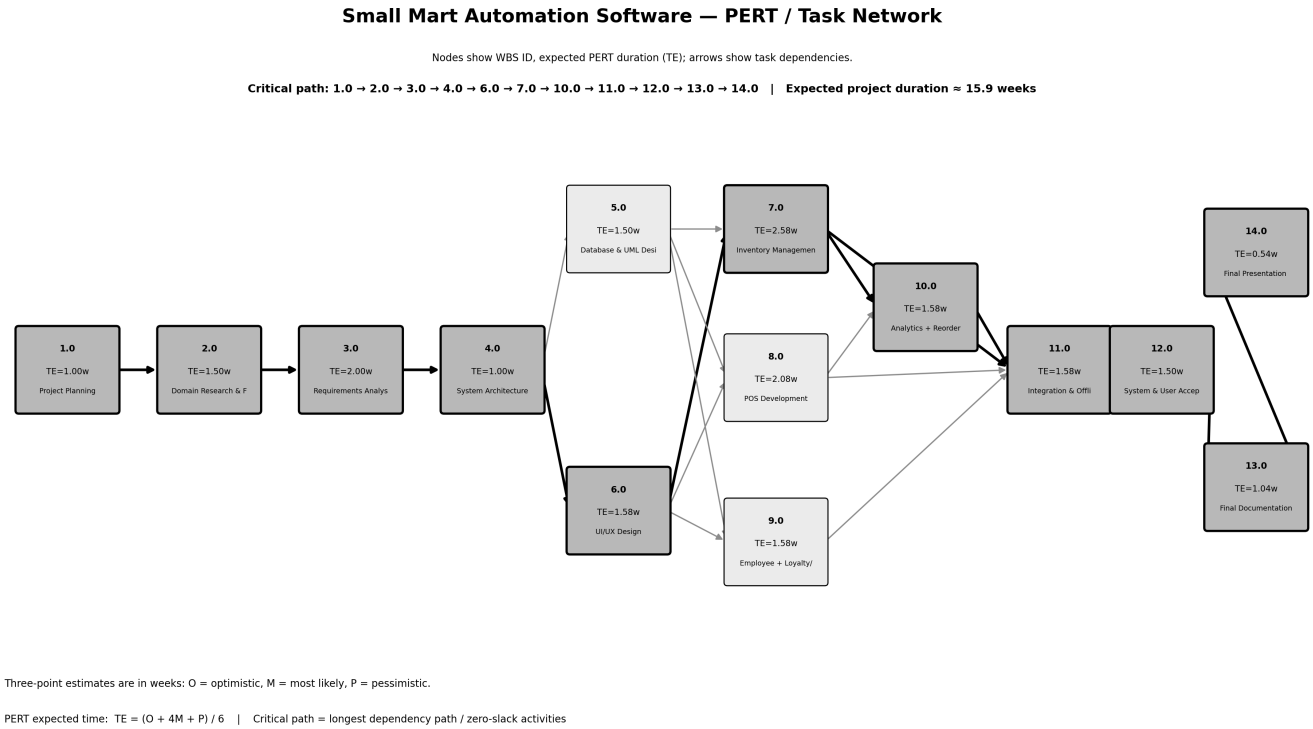


Figure 4: PERT/task network for the Small Mart Automation project. Arrows represent dependencies; critical-path activities are emphasized.

5.1 PERT Activity Estimates

WBS	Activity	Pred.	O	M	P	TE
1.0	Project Planning	—	0.50	1.00	1.50	1.00
2.0	Domain Research & Fieldwork	1.0	1.00	1.50	2.00	1.50
3.0	Requirements Analysis	2.0	1.00	2.00	3.00	2.00
4.0	System Architecture	3.0	0.50	1.00	1.50	1.00
5.0	Database & UML Design	4.0	1.00	1.50	2.00	1.50
6.0	UI/UX Design	4.0	1.00	1.50	2.50	1.58
7.0	Inventory Management	5.0, 6.0	1.50	2.50	4.00	2.58
8.0	POS Development	5.0, 6.0	1.50	2.00	3.00	2.08
9.0	Employee + Loyalty/CRM	5.0, 6.0	1.00	1.50	2.50	1.58
10.0	Analytics + Reordering + Loss Prevention	7.0, 8.0	1.00	1.50	2.50	1.58
11.0	Integration & Offline Capability	7.0, 8.0, 9.0, 10.0	1.00	1.50	2.50	1.58
12.0	System & User Acceptance Testing	11.0	1.00	1.50	2.00	1.50
13.0	Final Documentation	12.0	0.75	1.00	1.50	1.04
14.0	Final Presentation & Submission	13.0	0.25	0.50	1.00	0.54

Critical path: 1.0 → 2.0 → 3.0 → 4.0 → 6.0 → 7.0 → 10.0 → 11.0 → 12.0 → 13.0 → 14.0. The calculated expected project duration is approximately 15.9 weeks. Activities such as POS Development and Employee + Loyalty/CRM contain scheduling slack and can be adjusted without changing the planned project finish, provided their dependencies are respected.

6. Additional Information and Deliverable Checklist

The project should be developed incrementally. The team should maintain a shared backlog, agree on module interfaces before parallel development, and review the inventory model whenever a new feature changes stock state. The following checklist summarizes the expected first-stage project deliverables.

Deliverable	Status / action
Project scope and problem statement	Defined in this document.
Stakeholders and role descriptions	Defined; refine after fieldwork.
Domain fieldwork	Planned; insert interview findings after visit.
Competitive analysis	Initial comparison included; verify details during research.
Use case diagram	Create using the roles and use cases defined here.
Class diagram	Create from the proposed domain entities.
Sequence diagram	Create at least one for item scanning / sale processing.
Database schema / ERD	Create from Product, Inventory, Sale, Employee, Supplier and Customer entities.
Architecture diagram	Create cloud + offline synchronization architecture.
Gantt chart	Included; synchronized with PERT schedule.
PERT chart	Included; dependencies and critical path defined.
Prototype / implementation	To be developed according to the schedule.
Testing and evaluation	To be completed before final submission.

6.1 Recommended Development Order

1. Establish the product, inventory and employee data model. 2. Implement authentication and role-based access. 3. Build inventory receiving and stock adjustment workflows. 4. Build the POS flow around the inventory service. 5. Add analytics, reordering and loss-prevention rules. 6. Integrate offline synchronization. 7. Complete system and user acceptance testing. 8. Finalize documentation and presentation.

6.2 Scope Control

If time becomes limited, the team should preserve the core inventory + POS workflow and treat loyalty, self-checkout, BYOD QR shopping and advanced forecasting as extensions. A smaller, integrated and tested system is preferable to a large collection of incomplete modules.

End of Project Specification